Aqueous humour supplies nutrients and oxygen to the lens, cornea and some retinal cells. The aqueous humor is produced and drained at the same rate, maintaining a constant intra ocular pressure of about 16 mmHg. Any block in the canal of schlemm increases the infra ocular pressure of aqueous humor and leads to 'Glaucoma' where the optic nerve and the retina are compressed due to pressure.

10.6.2 Phonoreceptor

The ear is the site of reception of two senses namely hearing and equilibrium. Anatomically, the ear is divided into three regions: the external ear, the middle ear and internal ear**.**

The external ear consists of **pinna, external auditory meatus** and **ear drum**. The pinna is flap of elastic cartilage covered by skin. It collects the sound waves. The external auditory meatus is a curved tube that extends up to the tympanic membrane [the ear drum]. The tympanic membrane is composed of connective tissues covered with skin outside and with mucus membrane inside.

There are very fine hairs and wax producing sebaceous glands called **ceruminous glands** in the external auditory meatus. The combination of hair and the **ear wax** [cerumen] helps in preventing dust and foreign particles from entering the ear.

The middle ear is a small air-filled cavity in the temporal bone. It is separated from the external ear by the eardrum and from the internal ear by a thin bony partition; the bony partition contains two small membrane covered openings called the oval window and the round window.

The middle ear contains three ossicles: **malleus** [hammer bone], **incus** [anvil bone] and **stapes** [stirrup bone] which are attached to one another. The malleus is attached to the tympanic membrane and its head articulates with the incus which is the intermediate bone lying between the malleus and stapes. The stapes is attached to the oval window in the inner ear. The ear ossicles transmit sound waves to the inner ear. A tube called Eustachian tube connects the middle ear cavity with the pharynx. This tube helps in equalizing the pressure of air on either sides of the ear drum.

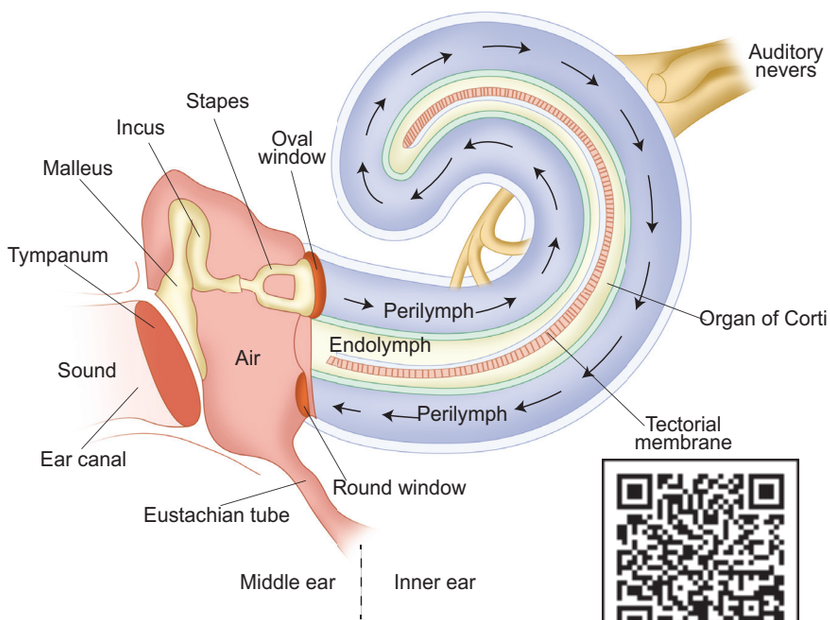


Figure 10.17 Path of Sound wave

Inner ear is the fluid filled cavity consisting of two parts, the bony labyrinth and the membranous labyrinths. The bony labyrinth consists of three areas: **cochlea, ve**stibule and semicircular canals. The cochlea is a coiled portion consisting of 3 chambers namely: scala vestibuli and scala tympani- these two are filled with perilymph; and the scala media is filled

TN_GOVT_XI_Zoology_Ch10.indd 193 09-01-2020 5.19.00 PM